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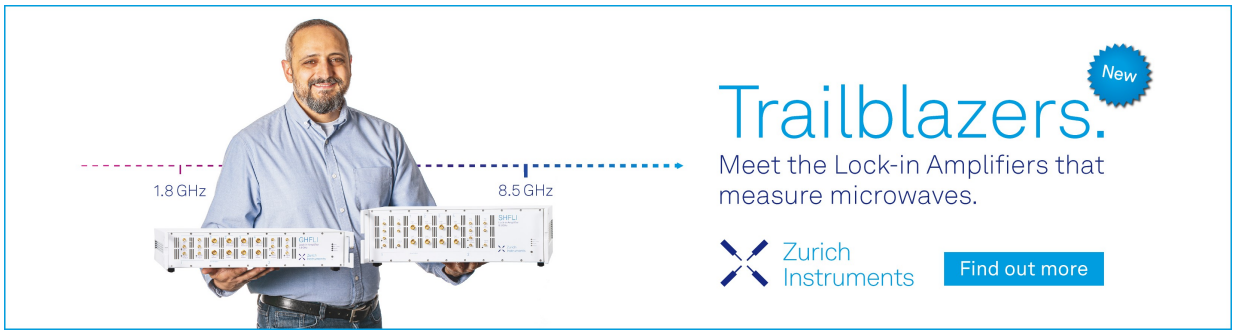

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Yield Modelling in Support of Industrial Investment Planning

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Abstract. Modelling and forecasting bond yields is a very important aspect in investment planning and making sound choices on different financing sources. In this paper we use time series of corporate and government bond yields in order to separate and study the impact of short- and long-term factors. Based on Nelson-Siegel ([1]) and Svensson ([2]) models, we analyse the dynamics of parameter estimates and use the results to explain market participants expectations. Unlike other studies, that aim at providing better estimates of the yield curve at specific moment in time, our major goal is to draw conclusions on the changes in the model estimates over time, as a proxy for different factors influencing bond yields.

INTRODUCTION

There are a number of important reasons for modelling bond yields and interest rates. Following the well-established practice of grouping yields by maturity, we can list several of these motives, that are particularly important for industrial investment planning:

- Assessment of interest rate risks needs to take into consideration differences in volatility of long- and short-term yields. A more detailed examples of macroeconomic outlook and volatility analysis can be found in [3], [4].

In order to transform theoretical models into a successful investment planning, it is however necessary not only to identify a suitable model and estimate its parameters, but to pick out what drives changes in bond yields and interest rates.

- The need for an early warning for potential adverse economic conditions becomes crucial when planning for a longer period. While we recognize that methods using real options analysis, as in [5] and [6], can successfully account for available flexibility in investment decisions, there is still a limit in terms of necessary time and costs in transforming industrial investment plans.

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If we consider that yield curve reflects the market expectations of the future interest rates, then observed curve can be used not only as a snapshot forecast, but also to judge on how investors foresee future, compared to current economic conditions.

- Planning for use of a specific source of financing can benefit from yield modelling as well. There are at least two ways to take advantage from such models – by choosing the right combination of bonds with different maturities and by taking advantage of investor preferences.

It should be noted that availability of the yield curve information does not mean that risk from changes in interest rates can be removed or anticipated with absolute precision. As demonstrated in [7], level, slope and curvature of the yield curve can be used as inputs for selecting the optimal maturity for both financing and investing. Therefore, it also makes sense for investors to monitor how these parameters change in order to constantly enhance and check their decision rules.

APPLICATION OF NELSON-SIEGEL-SVENSSON MODEL

It is possible to model yield curves with models of varying complexity – like simple polynomial, exponential or cubic splines, Nelson-Siegel and Nelson-Siegel-Svensson [8]. In this study we use the basic dependency of yield *r*, as a function of time to maturity *t*:

$$r(t) = \beta_0 + \beta_1 \left[\frac{1 - e^{-t/\gamma}}{t/\gamma}\right] + \beta_2 \left[\frac{1 - e^{-t/\gamma}}{t/\gamma} - e^{-t/\gamma}\right]. \quad (1)$$

Equation (1) has four parameters that have to be estimated and explained - $\beta_0, \beta_1, \beta_2$ and γ. The first three of them are of special interest for our study, as they are able to explain the impact of different groups of factors on observed yields. Table 1 provides a summary on the meaning of each parameter and how it can be used to support industrial investment planning.

TABLE 1. Nelson-Siegel-Svensson (NSS) parameters

	Role and meaning	Use with regard to investment planning
β_0	As it is not a function of the time to maturity, this parameter represents long-term expected returns.	Changes in this parameter would represent how long term expected returns rise or fall, therefore they can be used to check the validity of all assumptions related to discount factors for example.
β_1	Represents the short-term effects on yields.	Due to the exponential decay of the $\left[\frac{1-e^{-t/\gamma}}{t/\gamma}\right]$ term as time increases, the importance of this parameter will be negligible for large values of *t*. Therefore, changes in β_1 can help in estimating the impact of shorter-term financing options and also be used in assessment of risks related to sudden, but short living, changes in economic conditions.
β_2	Represents effects that are clearly visible in mid-term periods.	Due to the fact that the value of $\left[\frac{1-e^{-t/\gamma}}{t/\gamma} - e^{-t/\gamma}\right]$ also decreases with time, the impact of this term will also be lower. The rate at which it decreases depends also on γ and the effects of this term will last longer, compared to those of the previous term.

The estimated value of the last parameter – γ, indicates the relative importance of short- and mid-term effects on the yields. Example of the impact from changes in γ is shown on Figure 1. Changes of this parameter would indicate

that there is a shift in the weight that investors put on short- and mid-term factors when deciding on the required returns.

With regard to use of NSS parameters in practice, we can split the estimated values into the following groups:

- Values and estimates that are intrinsically important, thus the obtained results can be directly transformed into decision rules and can be applied into other analytical models.

Parameters $\beta_0, \beta_1, \beta_2$ are such examples, where each estimate can be used directly into taking decisions on use of specific source of financing, or applying the calculated yields in estimation of present value of expected cash flows.

- Values and estimates that are important, but even more significant is the way they change.

A notable example of such parameter would be γ, where the change in the importance of different factors proves even more valuable information, than the estimated value itself.

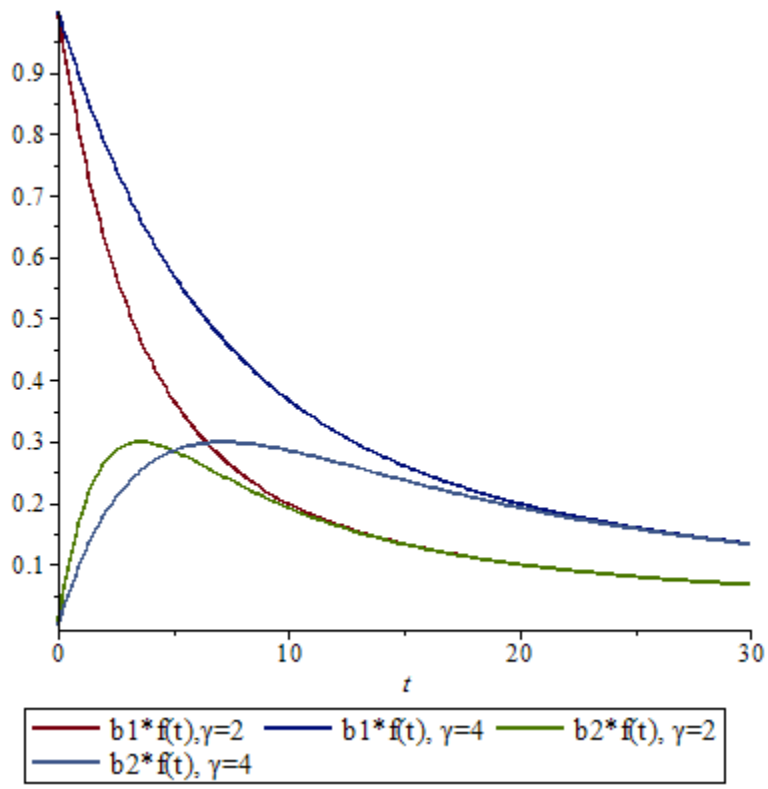

FIGURE 1. Plots of $\left[\frac{1-e^{-t/\gamma}}{t/\gamma}\right]$ (b1) and $\left[\frac{1-e^{-t/\gamma}}{t/\gamma} - e^{-t/\gamma}\right]$ (b2) under different assumptions for γ.

There exists an extension of the NSS model, as shown in equation (2), with additional parameters β_2 and γ_2. By introducing these parameters, it is possible to capture situations and yield curve effects that are not well explained by equation (1), at the expense of more demanding and more complex estimation procedure.

$$r(t) = \beta_0 + \beta_1\left[\frac{1 - e^{-t/\gamma_1}}{t/\gamma_1}\right] + \beta_2\left[\frac{1 - e^{-t/\gamma_1}}{t/\gamma_1} - e^{-t/\gamma_1}\right] + \beta_3\left[\frac{1 - e^{-t/\gamma_2}}{t/\gamma_2} - e^{-t/\gamma_2}\right]. \quad (2)$$

PARAMETER ESTIMATION AND INPUT DATA

It is possible to estimate NSS parameters in different ways, as described in [9], [10] and [11]. Recent studies of the model calibrated with negative yields, as in [12], have demonstrated that it can be used and still provide valuable output. Therefore, we have decided to use the variant shown by equation (2), and estimate all six parameters with a two-step procedure:

- define an interval for feasible values for γ_1 and γ_2.
- find optimal estimates for $\beta_0, \beta_1, \beta_2$ and β_3 in the already defined ranges for γ_1 and γ_2.

As an optimization criterion we use the squared residuals and the best fit is considered the set of parameters that produces the smallest sum of squared residuals. It is possible use a different metric for fitness of each solution, however our tests have shown that sum of squared residuals provides stable and accurate results.

Input Data Used

We use data that is published from Federal Reserve Bank of St. Louis[1] and covers securities with maturities ranging from 3 months till 30 years. Daily quotations and time series since the start of 1981 have been analyzed, which leads to a bit over 10 000 observations. A summary of the inputs is shown in Table 2.

TABLE 2. Summary of input time series, used to estimate NSS parameters and their evolution over time

	Security analysed	Information
DTB3	3-Month Treasury Bill: Secondary Market Rate	Min. :-0.050 1st Qu.: 0.910 Median : 3.800 Mean : 3.928 3rd Qu.: 5.700 Max. :17.010
DTB6	6-Month Treasury Bill: Secondary Market Rate	Min. : 0.020 1st Qu.: 1.000 Median : 3.940 Mean : 4.032 3rd Qu.: 5.900 Max. :15.930
DGS3	3-Year Treasury Constant Maturity Rate	Min. : 0.100 1st Qu.: 1.590 Median : 4.580 Mean : 4.922 3rd Qu.: 7.010 Max. :16.590
DGS5	5-Year Treasury Constant Maturity Rate	Min. : 0.190 1st Qu.: 2.220 Median : 4.860 Mean : 5.267 3rd Qu.: 7.330 Max. :16.270
DGS7	7-Year Treasury Constant Maturity Rate	Min. : 0.360 1st Qu.: 2.690 Median : 5.050

[1] https://fred.stlouisfed.org/series

Security analysed		Information
		Mean : 5.539 3rd Qu.: 7.540 Max. :16.050
DGS10	10-Year Treasury Constant Maturity Rate	Min. : 0.520 1st Qu.: 2.960 Median : 5.140 Mean : 5.734 3rd Qu.: 7.660 Max. :15.840
DGS20	20-Year Treasury Constant Maturity Rate	Min. : 0.870 1st Qu.: 3.660 Median : 5.690 Mean : 6.123 3rd Qu.: 7.850 Max. :15.780
DGS30	30-Year Treasury Constant Maturity Rate	Min. : 0.990 1st qu.: 3.840 Median: 5.560 Mean: 6.159 3rd Qu: 7.900 Max: 15.210

Data series plot, shown on Figure 2, reveals that there are periods of very low (close to zero) yields as well as times of negative values. This is a very good opportunity to test our approach under conditions which are not considered to be "a common case".

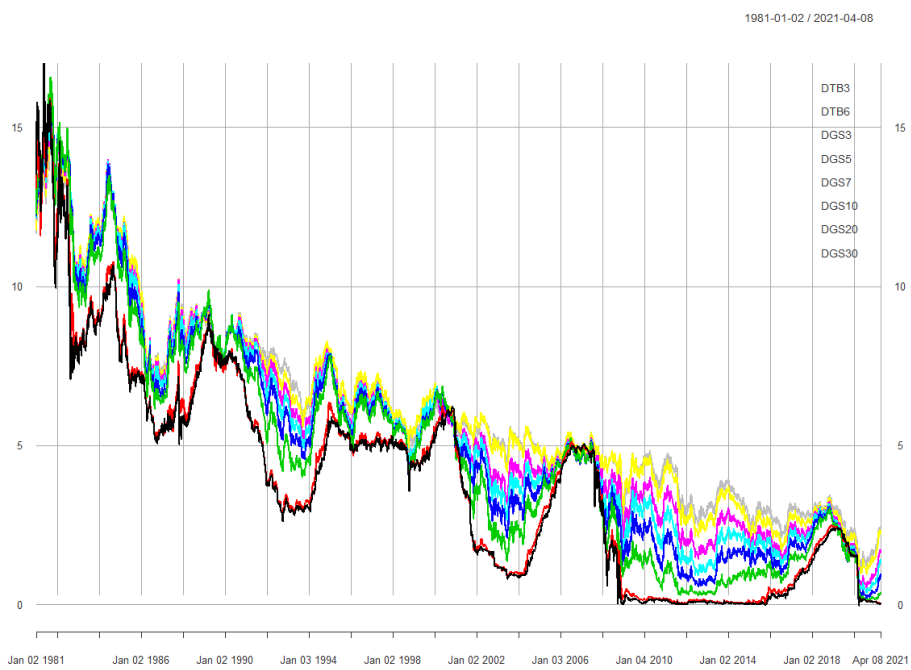

FIGURE 2. Plot of the input series over the complete study period, data provided by https://fred.stlouisfed.org/series

NUMERICAL RESULTS

Numerical results, presented in Table 3 are obtained by iterating through data set and estimating the NSS parameters at each point with the procedure described before. At each step, estimates are stored in separate time series, in order to process them later and access their characteristics. In order to compare how estimated parameters, change over the sample periods we use a special indicator, as described by equation (3).

$$V(\beta_i) = \frac{SD(\beta_i)}{MEAN(\beta_i)}, \tag{3}$$

where $\boldsymbol{SD(\beta_i)}$ is the standard deviation of $\boldsymbol{\beta_i}$ and $\boldsymbol{MEAN(\beta_i)}$ is the respective mean value, over the specified period.

In order to compare the performance of both NSS estimation and metric (3), we have examined how $\boldsymbol{V(\beta_i)}$ changes over three time periods – the first one covering the entire data set (e.g. from the beginning of 1981 till 08. April.2021), the second one including the peak of the financial crisis from 2008 and the third one focusing on the COVID-19 crisis and 2020-2021.

TABLE 3. Changes in NSS parameters over different periods

	Parameters	Standard deviation / Mean
1981 - 2021	β_0	0.9735726
	β_1	-17.76237
	β_2	-4.30335
	β_3	2.028569
2008-2009	β_0	28.18738
	β_1	-5.918754
	β_2	0.7313869
	β_3	0.3783354
2020-2021	β_0	0.1331955
	β_1	-0.2407084
	β_2	-1.621595
	β_3	-0.6657732

As Figure 3 shows, changes in the parameters can be used to reach better assessment of the economic situation and what drives investors in making specific decisions. Compared to simple analysis of the yield changes, it is possible stress on two important aspects – first to identify what is the key factor driving a particular change (for example new long-term expectations or just a short-term effect), and second to forecast only some changes in investor behavior.

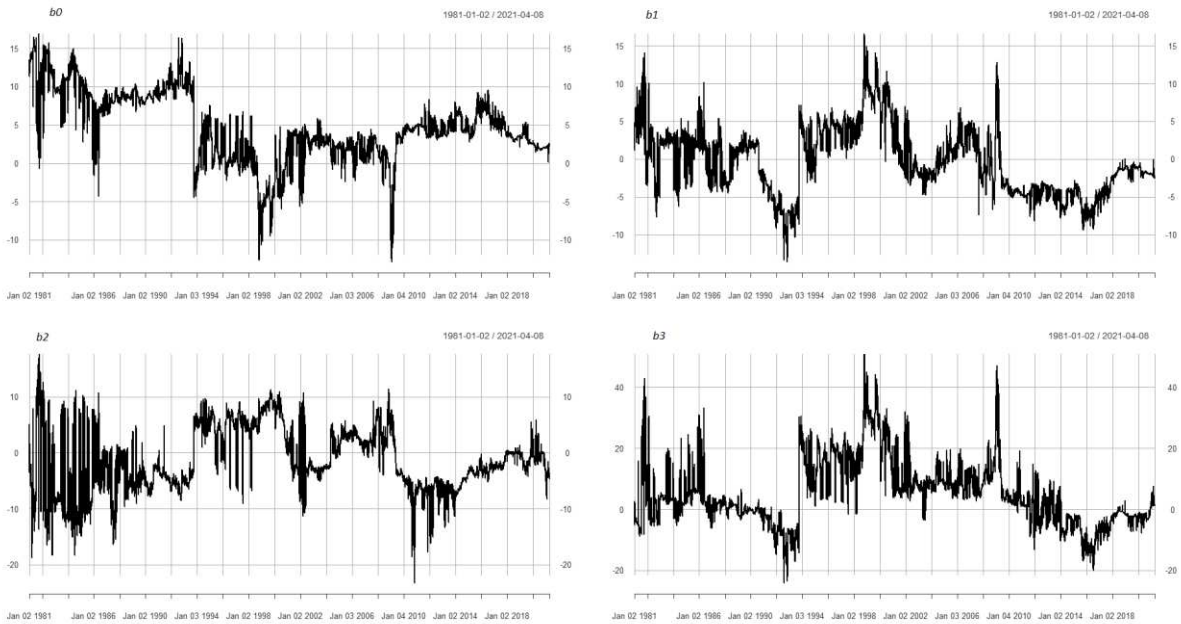

FIGURE 3. Changes in NSS estimates over time, using the entire dataset

With regard to industrial investments, results obtained on Figure 3 can be used to improve planning in the following areas:

- dynamics of the long-term expectations are important indicator for detecting changes in the environment and reflecting those changes in the investment plan.

For example, rise in the β_1 estimate would indicate that the importance of short-term factors has increased, thus it may be better to plan for financing options with a longer maturity.

- running forecasts on specific parameters can be used as a complementary tool in risk assessment.

Depending on the specific risks that need to be considered in the investment plan, making estimates that use volatility in one of the NSS parameters can give a better overview of the actual risk exposure.

- a comparison of the estimated parameters can be done, when calibrating NSS model against yields of fixed income securities in different countries, in order to plan for better allocation of investment funds.

CONCLUSIONS

Analysis of yield curves is a common and widely used tool, when studying the behavior of interest rates and fixed income securities. However, it can also be a very useful tool in support of investment planning for industrial enterprises. Because of the time necessary to implement and fully utilize investments in real assets, it is often necessary to plan in details what will be the impact of different changes in interest rates and bond prices. Such a process if crucial regardless if companies want to issue bonds or just invest some of the available funds.

We have used NSS model to study changes in the yield curve and focus on changes in parameter estimates as a tool for understanding market dynamics and investor's concerns. To test if such an approach is feasible we have tested it on publicly available FED data for securities that have a lifetime from 3 months up to 30 years. Our findings demonstrate, that studying changes in the estimates can be equally important, as the estimated values themselves, as it helps figure out how the importance of different factors evolves. Practitioners can use the results of this study as a support tool during their investment decision-making process. Different parts of the planning process can be improved – from risk assessment to discount factor estimation.

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